

COMPUTER NETWORKS

Week-2

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The Internet: a “nuts and bolts” view



Billions of connected computing **devices**:

- **hosts** = end systems
- running **network apps** at Internet's “edge”



Packet switches: forward packets (chunks of data)

- **routers, switches**

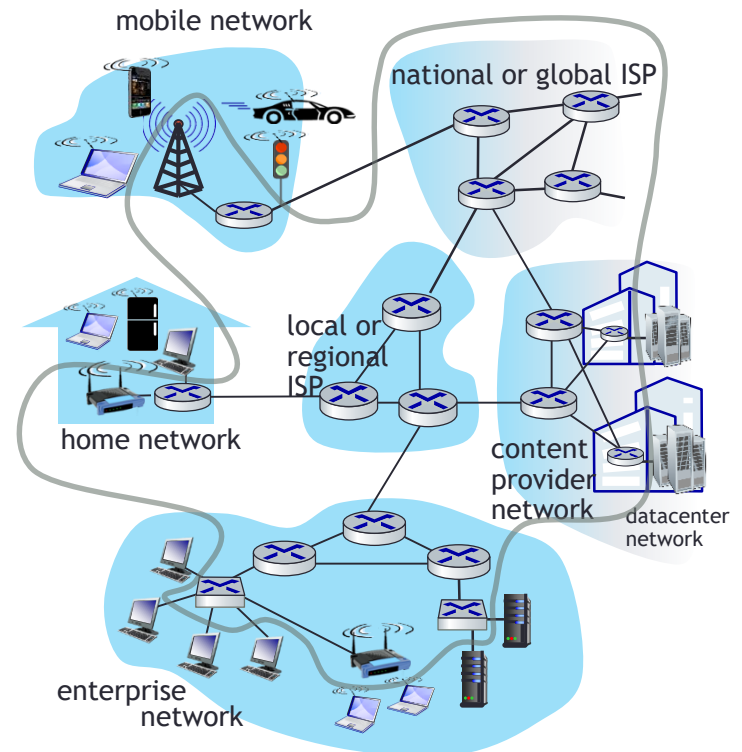


Communication links

- fiber, copper, radio, satellite
- transmission rate: **bandwidth**

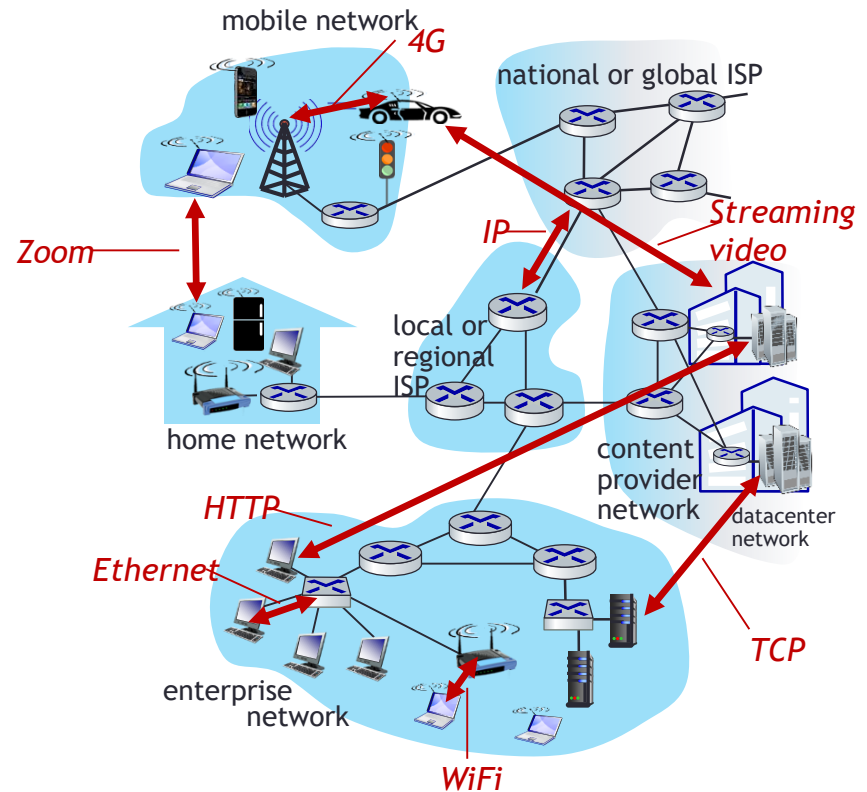
Networks

- collection of devices, routers, links: managed by an organization



The Internet: a “nuts and bolts” view

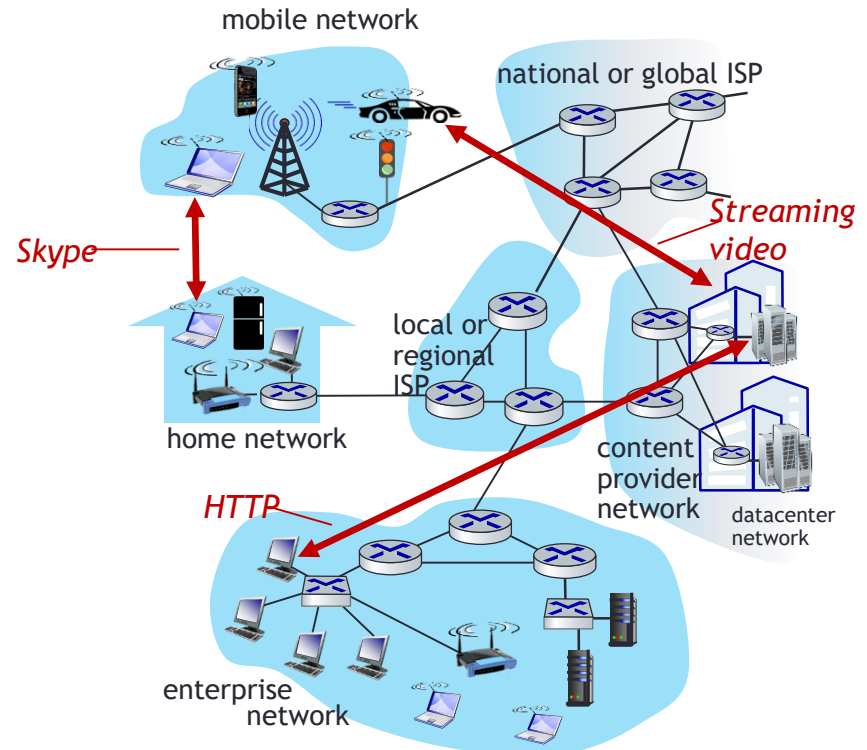
- *Internet*: “network of networks”
 - Interconnected ISPs
- *protocols* are everywhere
 - control sending, receiving of messages
 - e.g., HTTP (Web), streaming video, Skype, TCP, IP, WiFi, 4G, Ethernet
- *Internet standards*
 - RFC: Request for Comments
 - IETF: Internet Engineering Task Force



Example of content delivery network: **Netflix, YouTube, and Spotify**

The Internet: a “services” view

- *Infrastructure* that provides services to applications:
 - Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, ...
- provides *programming interface* to distributed applications:
 - “hooks” allowing sending/receiving apps to “connect” to, use Internet transport service
 - provides service options, analogous to postal service



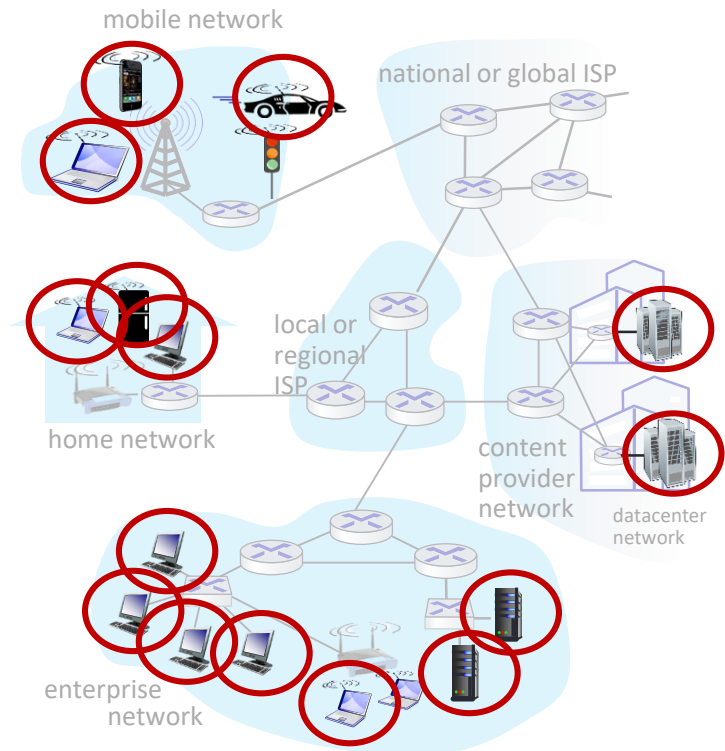
What's a protocol?

Protocols define the format, order of messages sent and received among network entities, and actions taken on message transmission, receipt

A closer look at Internet structure

Network edge:

- hosts: clients and servers
- servers often in data centers



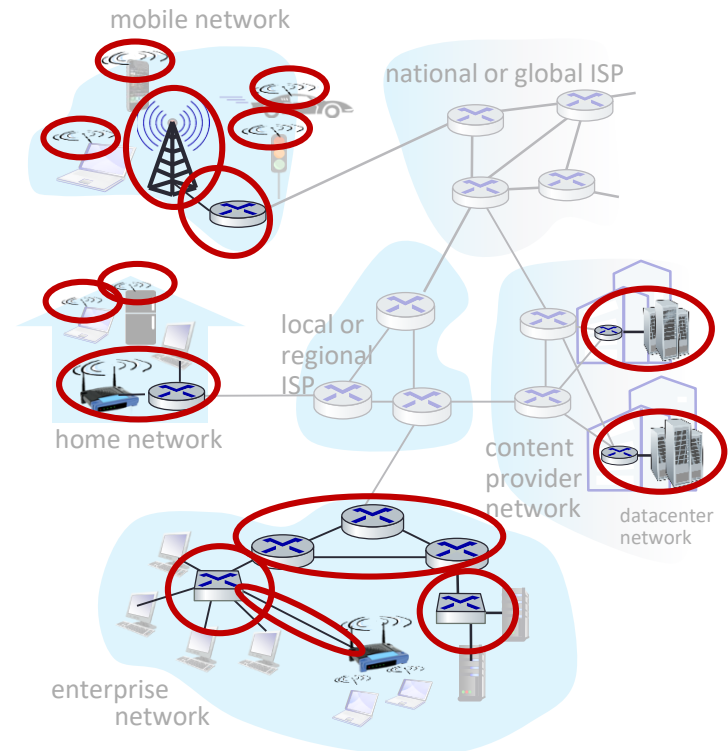
A closer look at Internet structure

Network edge:

- hosts: clients and servers
- servers often in data centers

Access networks, physical media:

- wired, wireless communication links



A closer look at Internet structure

Network edge:

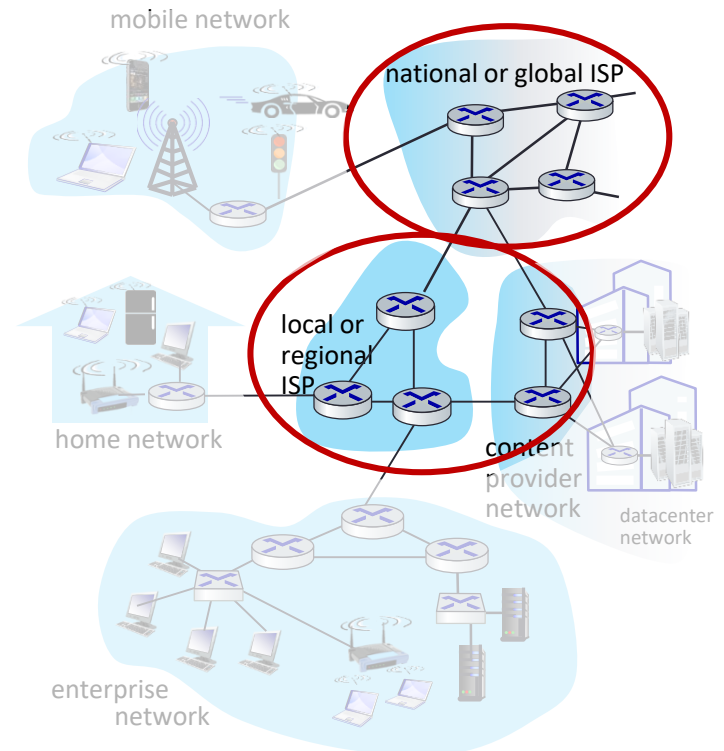
- hosts: clients and servers
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Access networks, physical media:

- wired, wireless communication links

Network core:

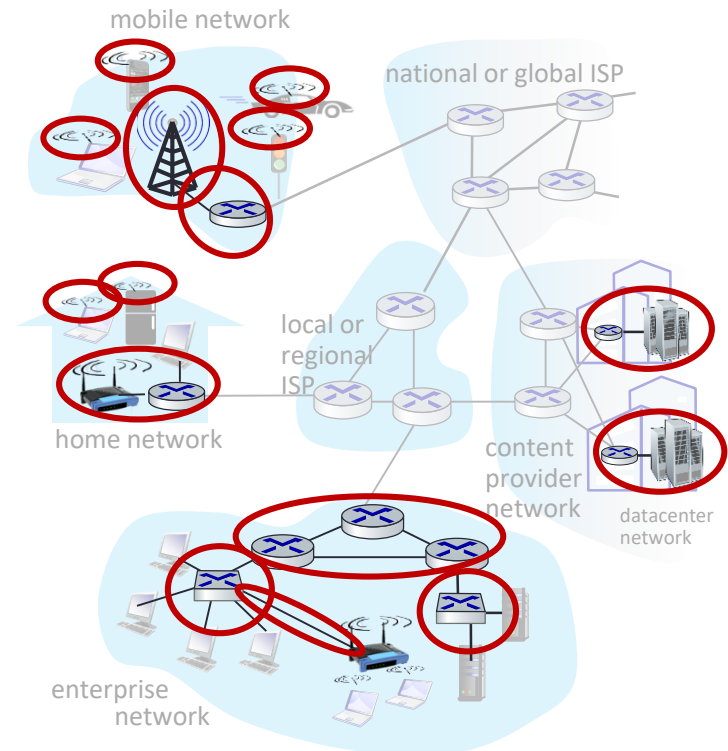
- interconnected routers
- network of networks



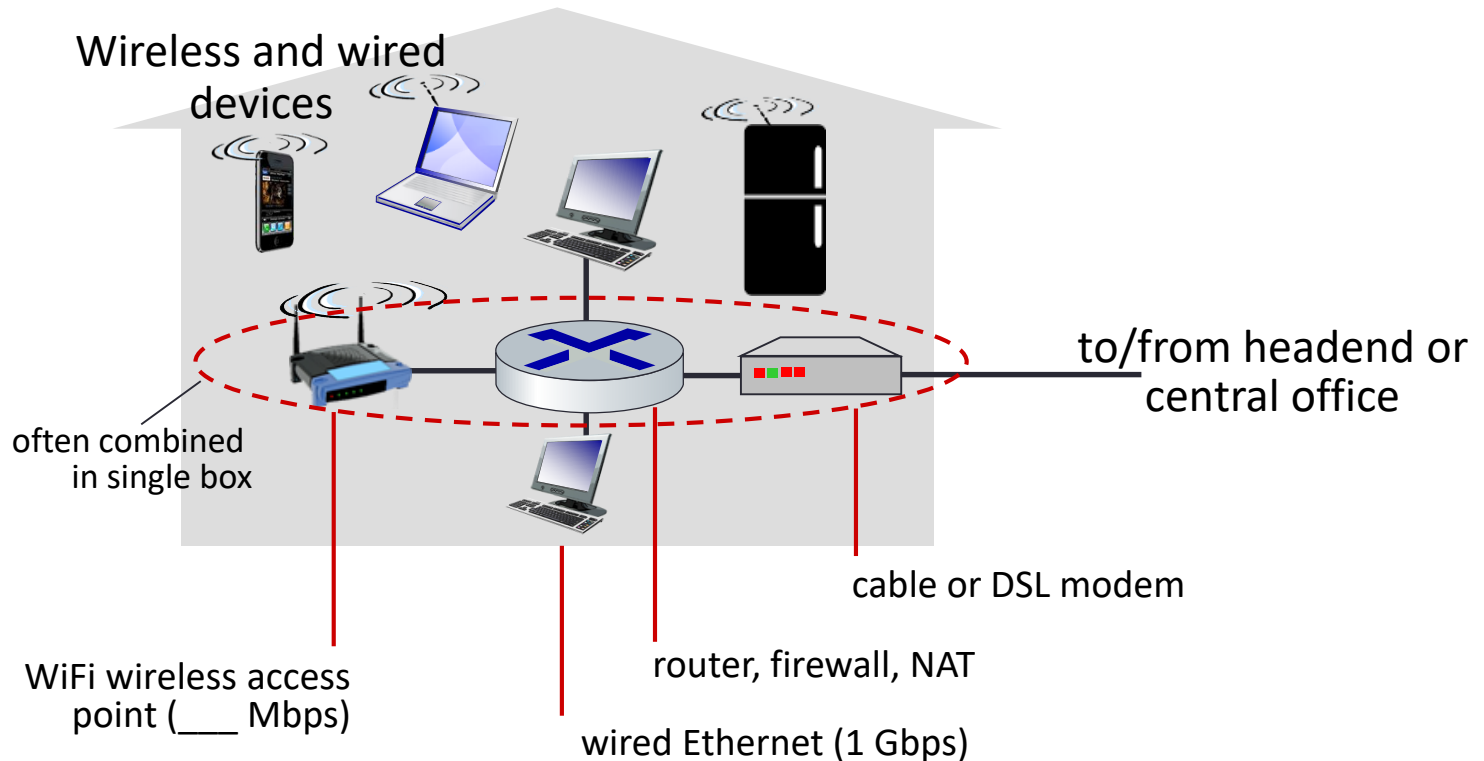
Access networks and physical media

Q: How to connect end systems to edge router?

- residential access nets
- institutional access networks (school, company)
- mobile access networks (WiFi, 4G/5G)



Access networks: home networks



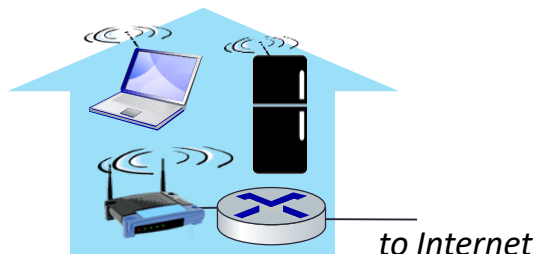
Wireless access networks

Shared *wireless* access network connects end system to router

- via base station aka “access point”

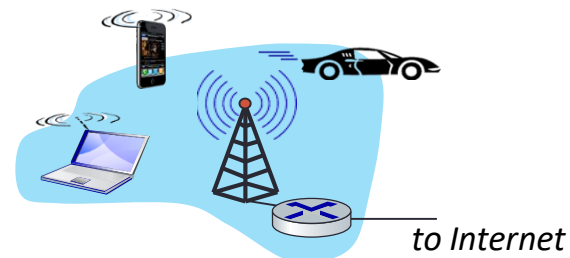
Wireless local area networks (WLANs)

- typically within or around building (~100 ft)

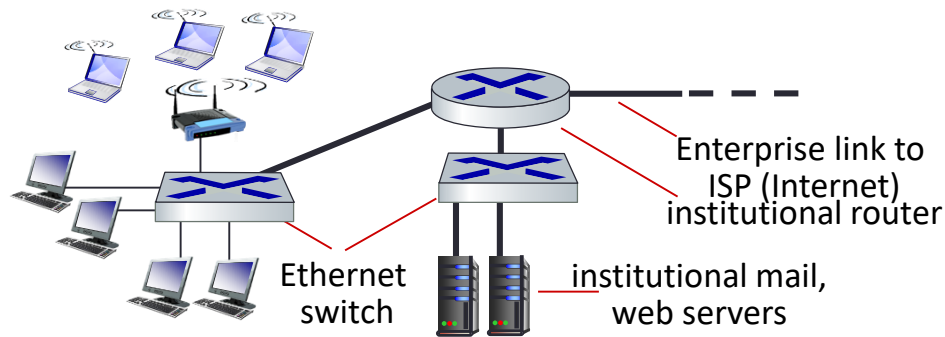


Wide-area cellular access networks

- provided by mobile, cellular network operator (10's km)
- 10's Mbps
- 4G cellular networks (5G coming)



Access networks: enterprise networks



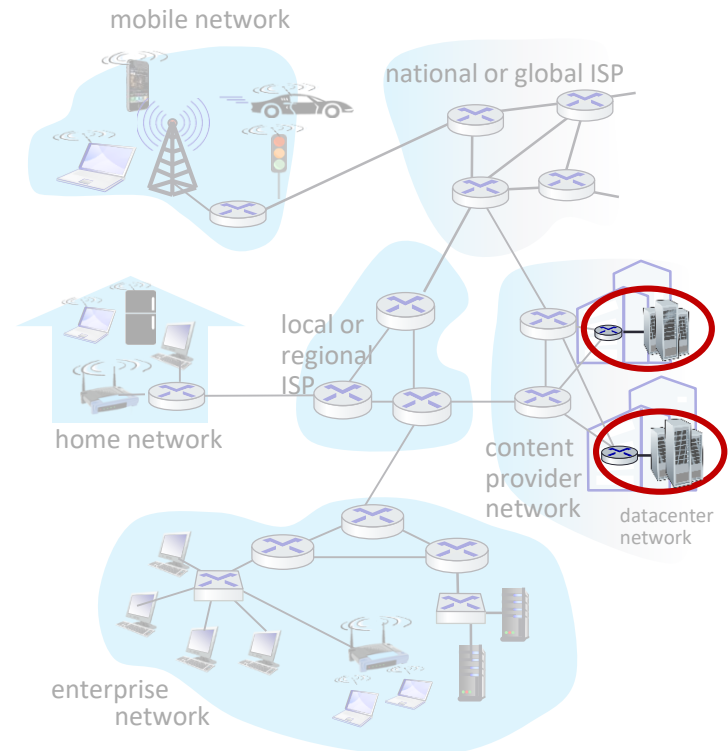
- companies, universities, etc.
- mix of wired, wireless link technologies, connecting a mix of switches and routers
 - Ethernet: wired access at 100Mbps, 1Gbps, 10Gbps
 - WiFi: wireless access points at 11, 54, 450 Mbps

Access networks: data center networks

- high-bandwidth links (10s to 100s Gbps) connect hundreds to thousands of servers together, and to Internet



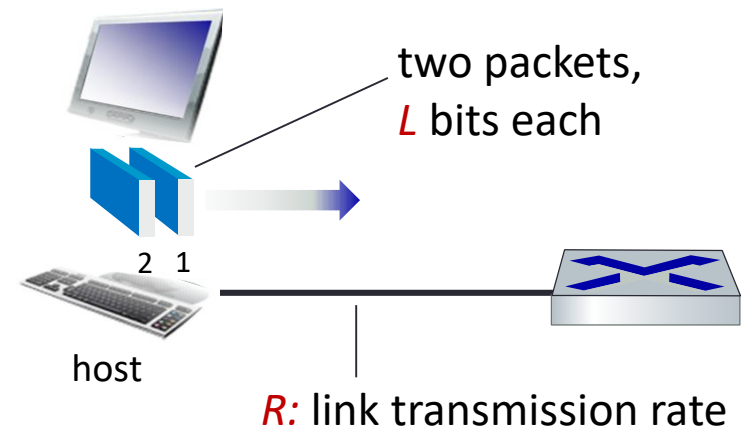
Courtesy: Massachusetts Green High Performance Computing Center (mghpcc.org)



Host: sends *packets* of data

host sending function:

- takes application message
- breaks into smaller chunks, known as *packets*, of length L bits
- transmits packet into access network at *transmission rate R*
 - link transmission rate, aka link *capacity, aka link bandwidth*



$$\text{packet transmission delay} = \text{time needed to transmit } L\text{-bit packet into link} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$

Transmission Media

Transmission medium

- the physical path between transmitter and receiver
- Repeaters or amplifiers may be used to extend the length of the medium

Transmission Media Tpyes

- By type of path:
 - Communication can follow an exact path e.g. wire
 - Communication can have no specific path, such as a radio transmission
- By form of energy:
 - Electrical energy - used on wires
 - Radio transmission - used for wireless
 - Light - used for optical fiber

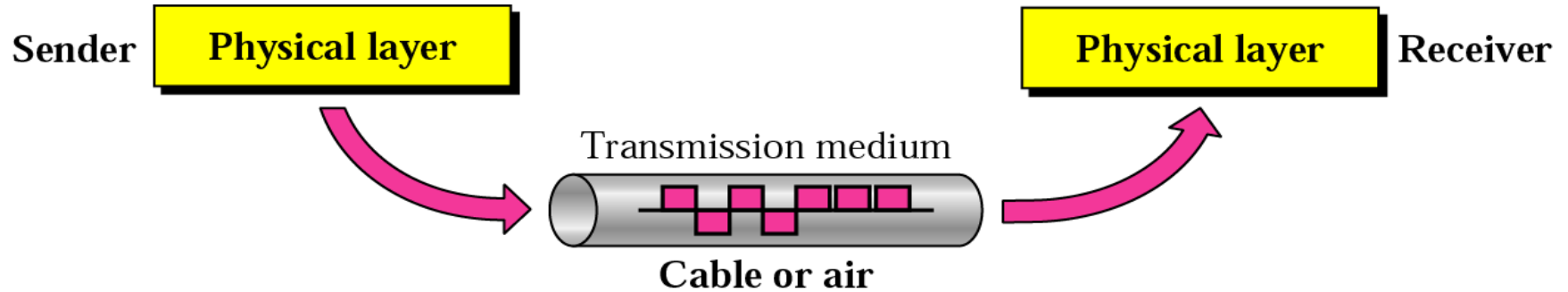
Transmission Media

- Communication of electromagnetic waves is *guided* or *unguided*.

Guided media : waves are guided along a physical path
(e.g, twisted pair, coaxial cable and optical fiber)

Unguided media: means for transmitting but not guiding
electromagnetic waves (e.g., the atmosphere and outer
space)

Physical media



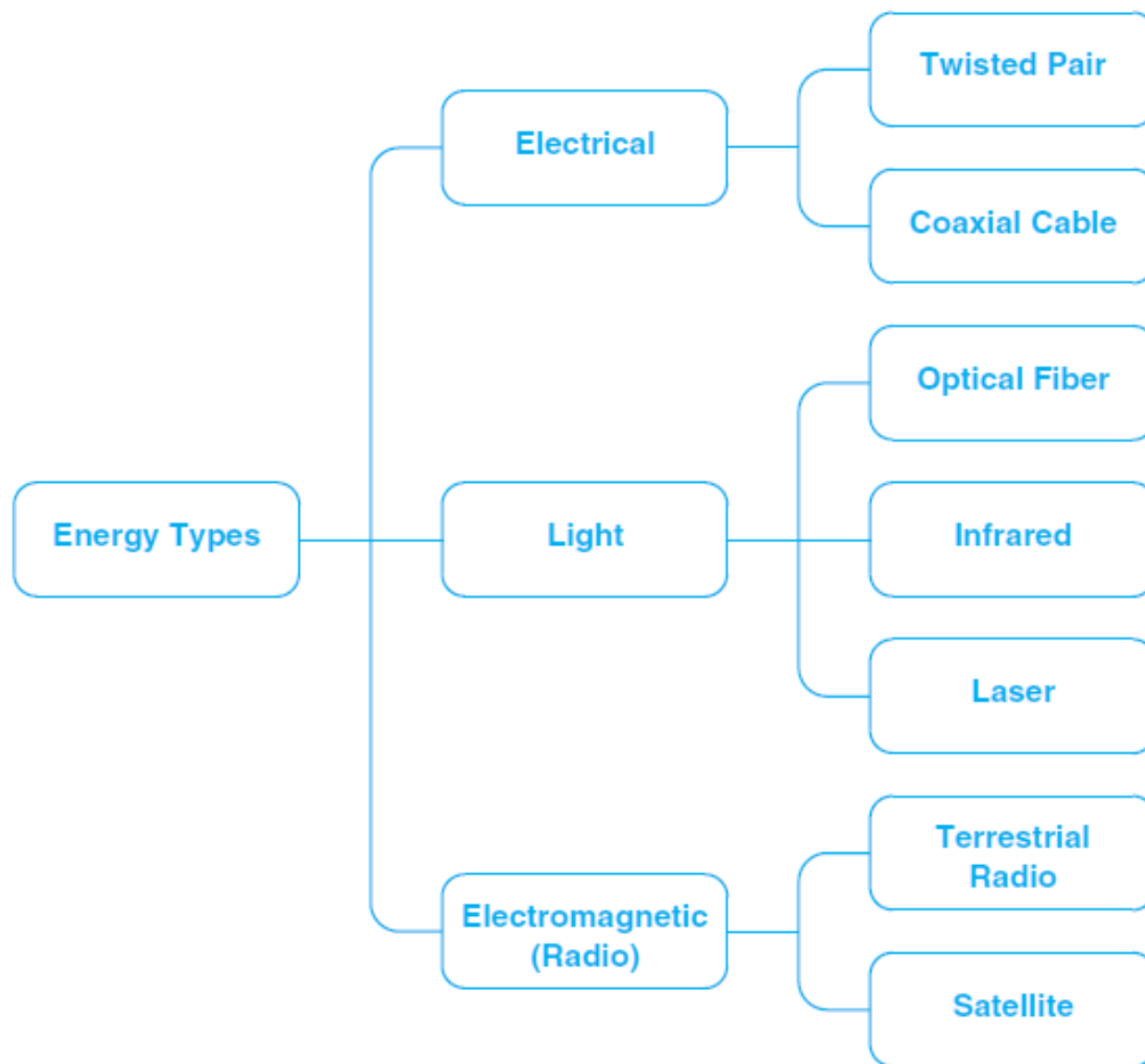
Repeater

- A WiFi repeater or extender is used to extend the coverage area of your WiFi network

Transmission Media Choices

- Twisted pair
- Coaxial cable
- Optical fiber
- Wireless communications

Find out bit rate for each transmission media

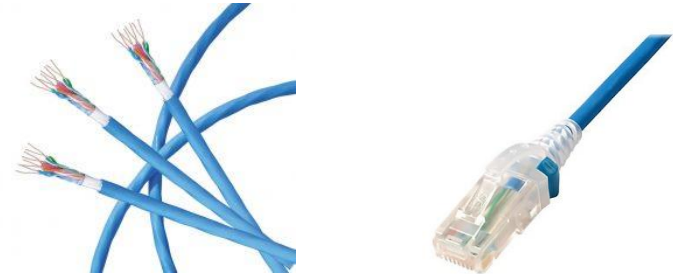


Links: Summary

- **bit:** propagates between transmitter/receiver pairs
- **physical link:** what lies between transmitter & receiver
- **guided media:**
 - signals propagate in solid media: copper, fiber, coax
- **unguided media:**
 - signals propagate freely, e.g., radio

Twisted pair (TP)

- two insulated copper wires
 - Category 5: 100 Mbps, 1 Gbps Ethernet
 - Category 6: 10Gbps Ethernet



Links:Summary

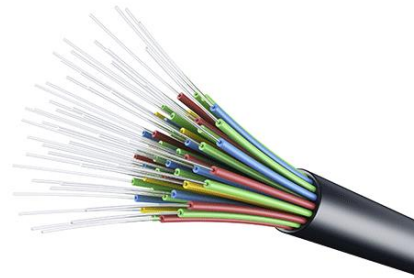
Coaxial cable:

- two concentric copper conductors
- bidirectional
- broadband:
 - multiple frequency channels on cable
 - 100's Mbps per channel



Fiber optic cable:

- glass fiber carrying light pulses, each pulse a bit
- high-speed operation:
 - high-speed point-to-point transmission (10's-100's Gbps)
- low error rate:
 - repeaters spaced far apart
 - immune to electromagnetic noise



Links: Summary

Wireless radio

- signal carried in various “bands” in electromagnetic spectrum
- no physical “wire”
- broadcast, “half-duplex” (sender to receiver)
- propagation environment effects:
 - reflection
 - obstruction by objects
 - Interference/noise

Radio link types:

- **Wireless LAN (WiFi)**
 - 10-100's Mbps; 10's of meters
- **wide-area (e.g., 4G cellular)**
 - 10's Mbps over ~10 Km
- **Bluetooth:** cable replacement
 - short distances, limited rates
- **terrestrial microwave**
 - point-to-point; 45 Mbps channels
- **satellite**
 - up to 45 Mbps per channel
 - 270 msec end-end delay

Design of a WAN

- Includes large number of Switching Nodes
- Transmission from any one device is routed through these internal nodes to the destination device
- These nodes are not concerned with the content of data
- Their purpose is to provide a switching facility that will move the data from node to node until it reaches its destination

WAN Implementation

- Traditionally WANs are implemented using one of two technologies
- Circuit Switching
- Packet Switching

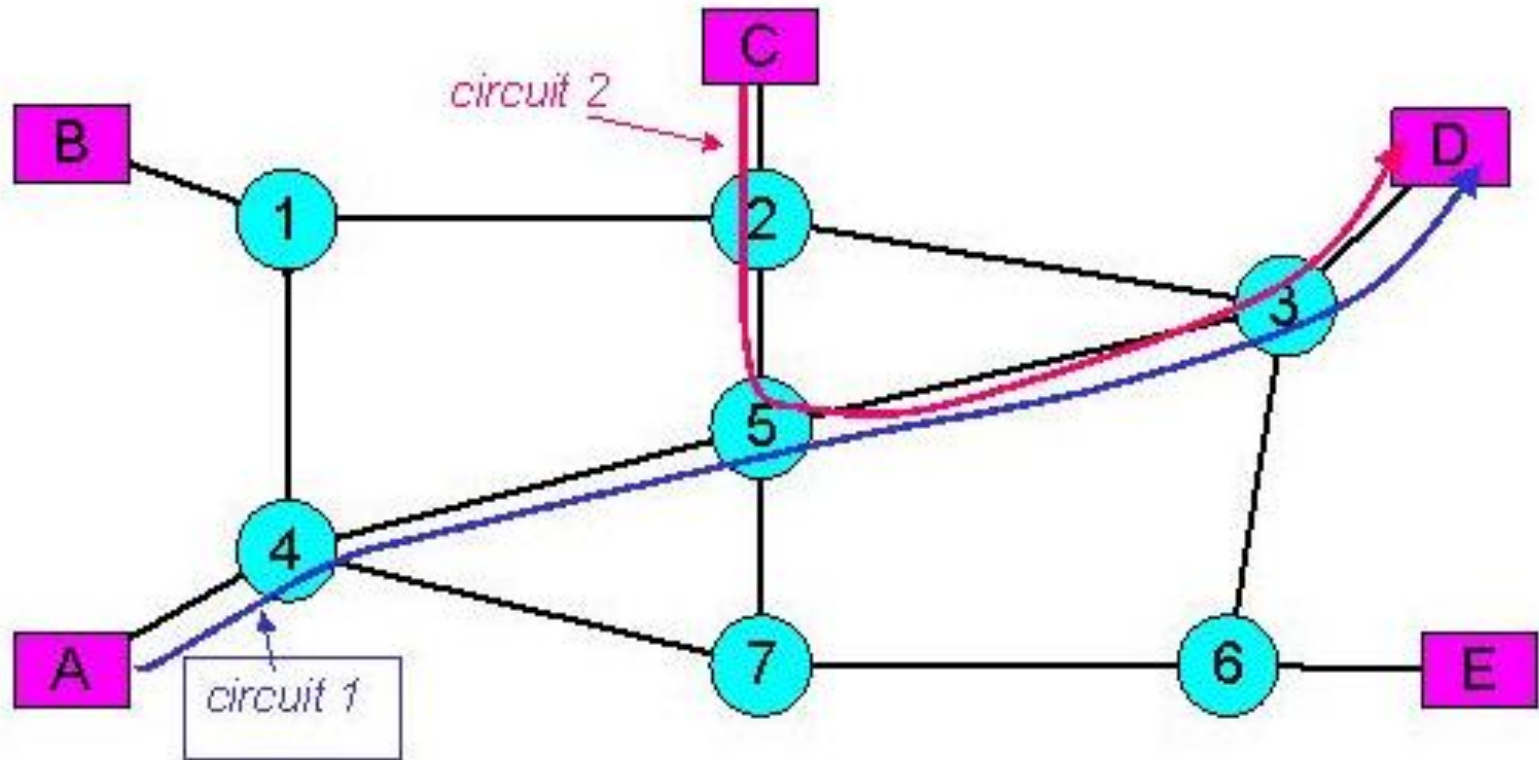
Circuit Switching

- Used for voice communication
- A dedicated communication path is established between two stations through the nodes of the network
- The path/CAPACITY stays up for the duration of communication/connection
- If there is no data, capacity is wasted
- Three phases
 - Establish
 - Transfer
 - Disconnect
- Example: Telephone Network

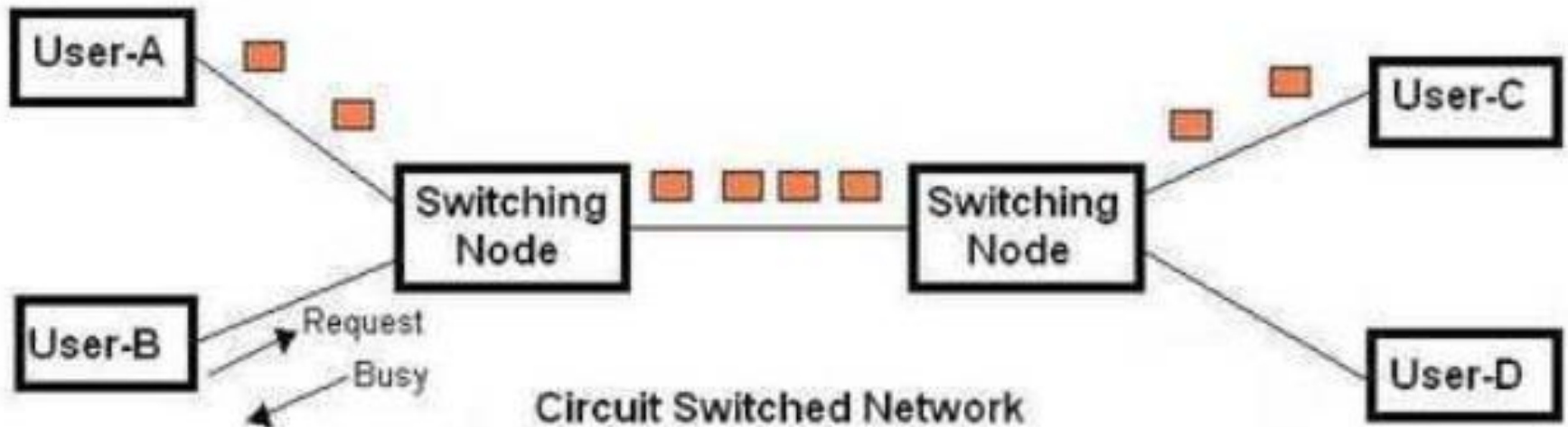
Circuit Switching

- Dedicated channel has to be established before the call is made between users
- The channel is reserved between the users till the connection is active
- For half duplex communication, one channel is allocated
- For full duplex communication, two channels are allocated
- It is mainly used for voice communication requiring real time services without any delay

Circuit Switching



Circuit Switching



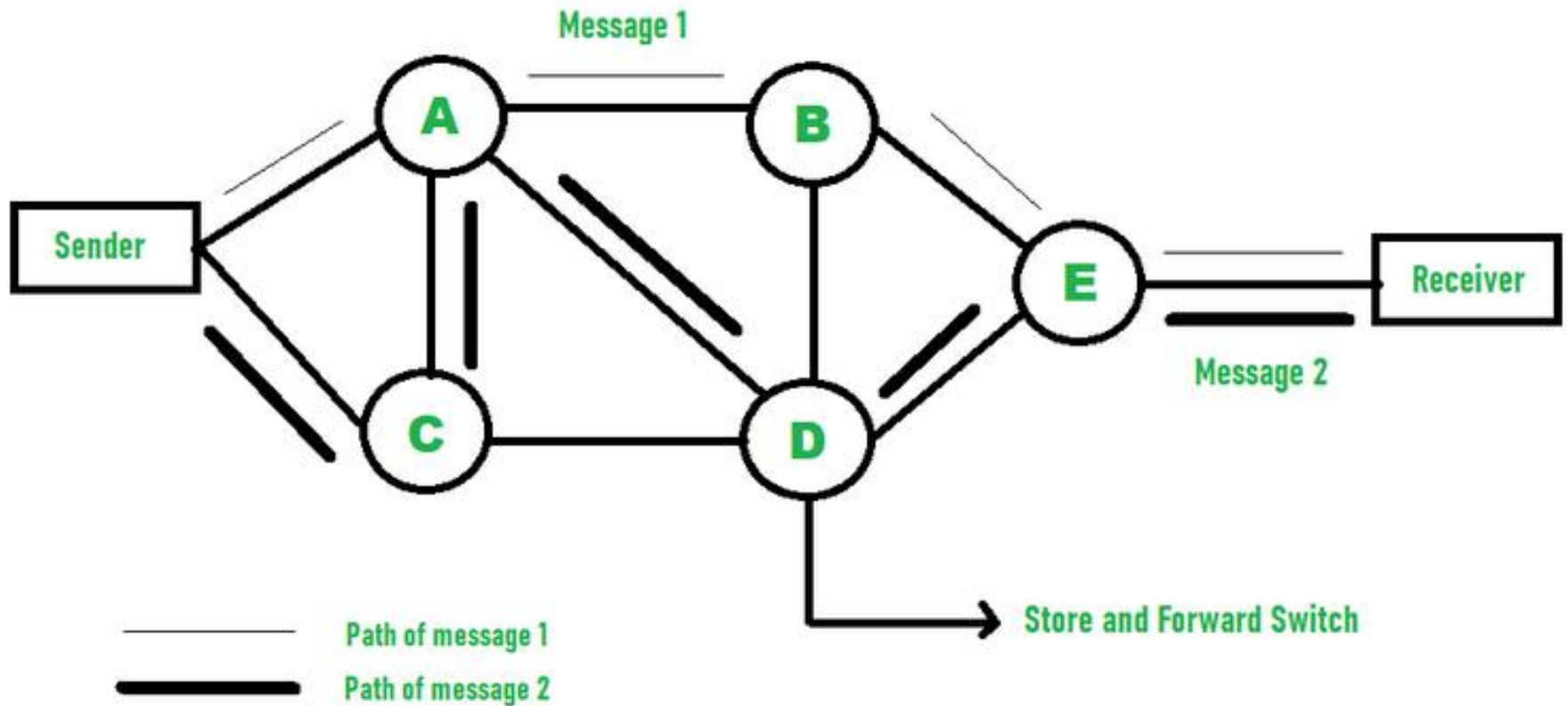
Circuit Switching

- if user-A wants to use the network to communicate with user-C, it needs to request a dedicated channel
- Now user-A can communicate with user-C
- During the connection phase if user-B tries to call/communicate with user-D or any other user it will get busy signal from the network

Message Switching

- It is used for message communication such as SMS, MMS etc.
- Complete message is transmitted without any break
- No limit on the block size
- Message is stored and transmitted at each node → more time required
- At a particular time, a message exists at one location in the network
- No direct link is available between sender and receiver
- Physical links are allocated dynamically

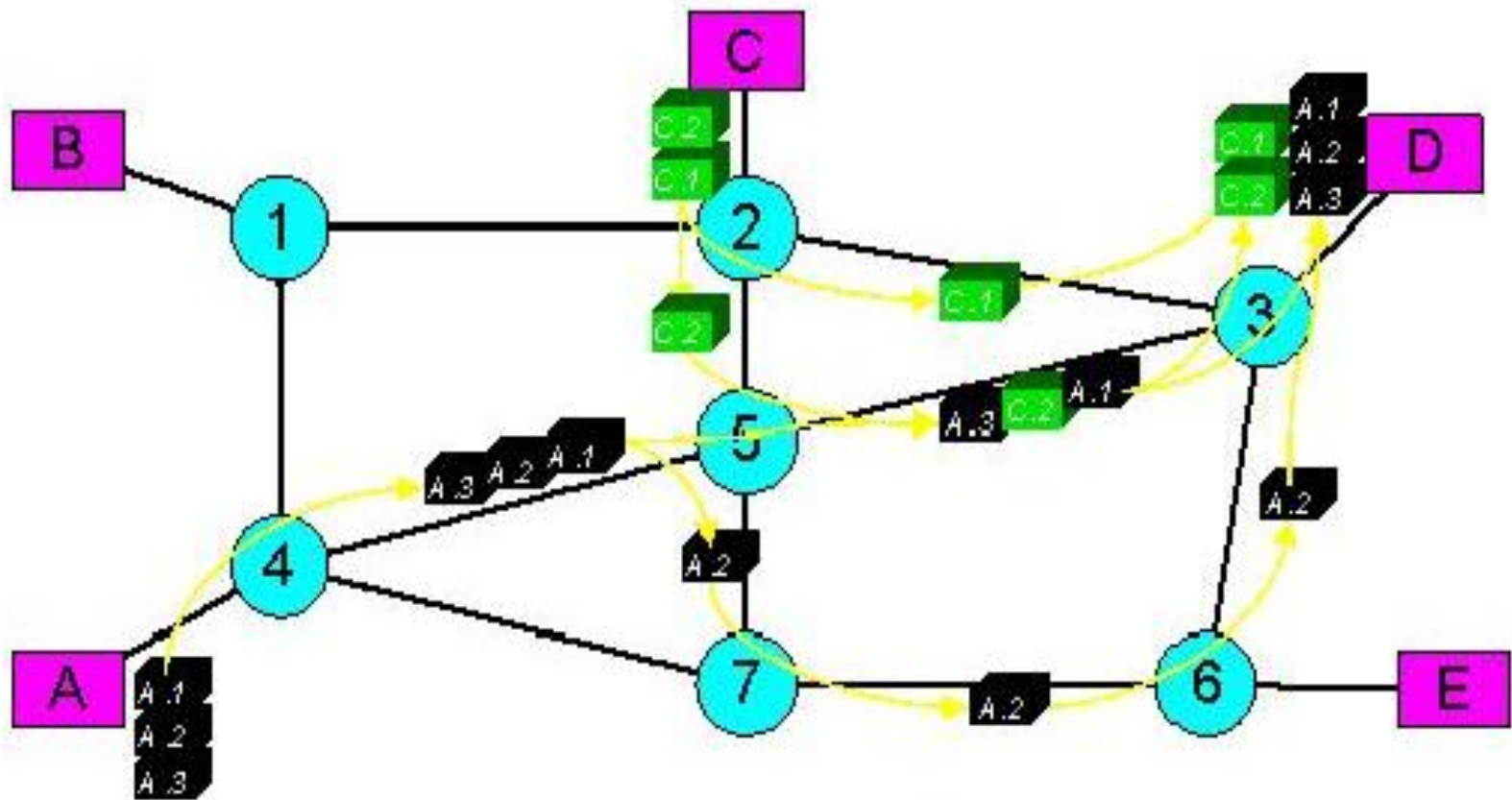
Message Switching



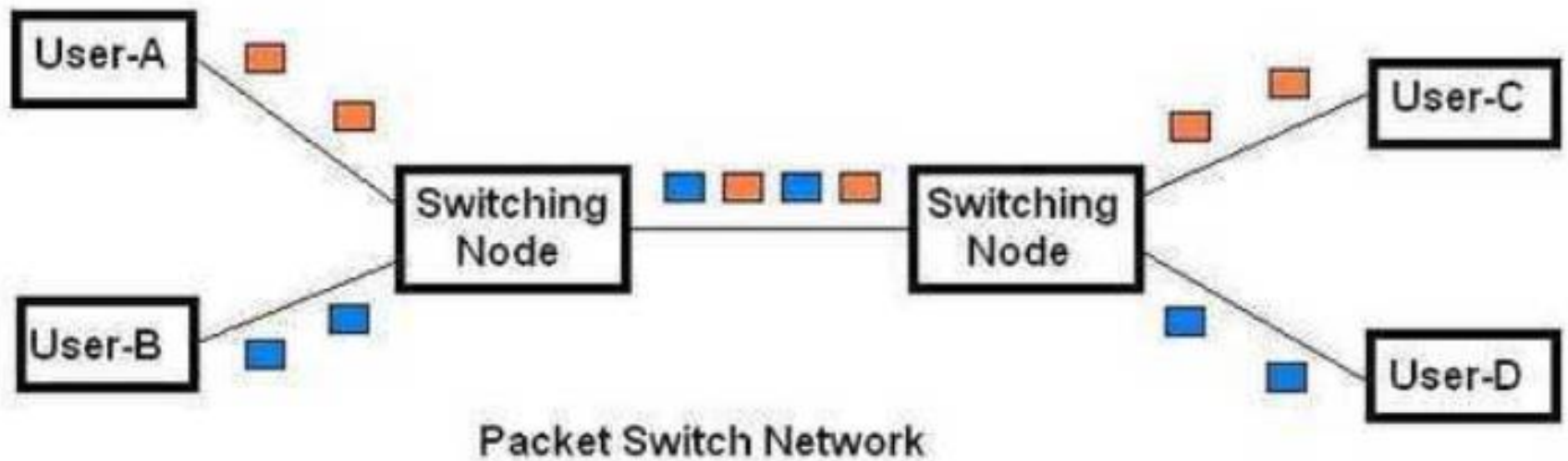
Packet Switching

- Used for Data Communication
- Does not require dedicated connection. It is useful for less time sensitive internet data or emails
- Data is sent out in small chunks called “Packets”
- Each packet is passed from node to node
- At each node, entire packet is received, stored briefly and then transmitted to the next node
- Example:
- Computer to Computer Communication

Packet Switching



Packet Switching



Packet Switching

- User-A wants to send data/information to user-C
- User-B wants to send data to user-D
- It is simultaneously possible
- Information is padded with header which contains addresses of source and destination
- This header is sniffed by intermediate switching nodes to determine their route and destination

Packet Switching

- Packet switching is similar to message switching using short messages
- Any message exceeding a network-defined maximum length is broken up into shorter units, known as Packets
→ upper limit on the block size
- Each packet is then transmitted individually through the virtual links that are made simultaneously
- So packets exist at many places in the network at any particular time

Packet Switching

- The performance of Packet Switching is called Best Effort performance
- If you transmit from sender to receiver, the network will do its best to get the packet to the other end as fast as possible
- → no guarantees on how fast that packet will arrive

Comparison

- Message switching takes more time compared to packet switching as entire message will be stored at each of the hop points till it is completely received

A Packet

- All the packets are sent with a 'header address' which tell about the address of its final destination
- The header address also illustrates the sequence for reassembly at the destination computer → the packets are rearranged into the correct order
- One packet also contains details of how many packets should be arriving so that the recipient computer knows if any packet has failed to arrive
- If a packet fails to arrive, the recipient computer sends a message back to the sender's computer, asking for the missing packet to be resent

Internet

- The Internet uses IP (Internet Protocol), which is a packet switching protocol
- The Internet uses packet switching, not circuit switching
- Circuit switching is very wasteful of resources because it dedicates an end-to-end circuit to a call, where packet switching can use a circuit for different communications

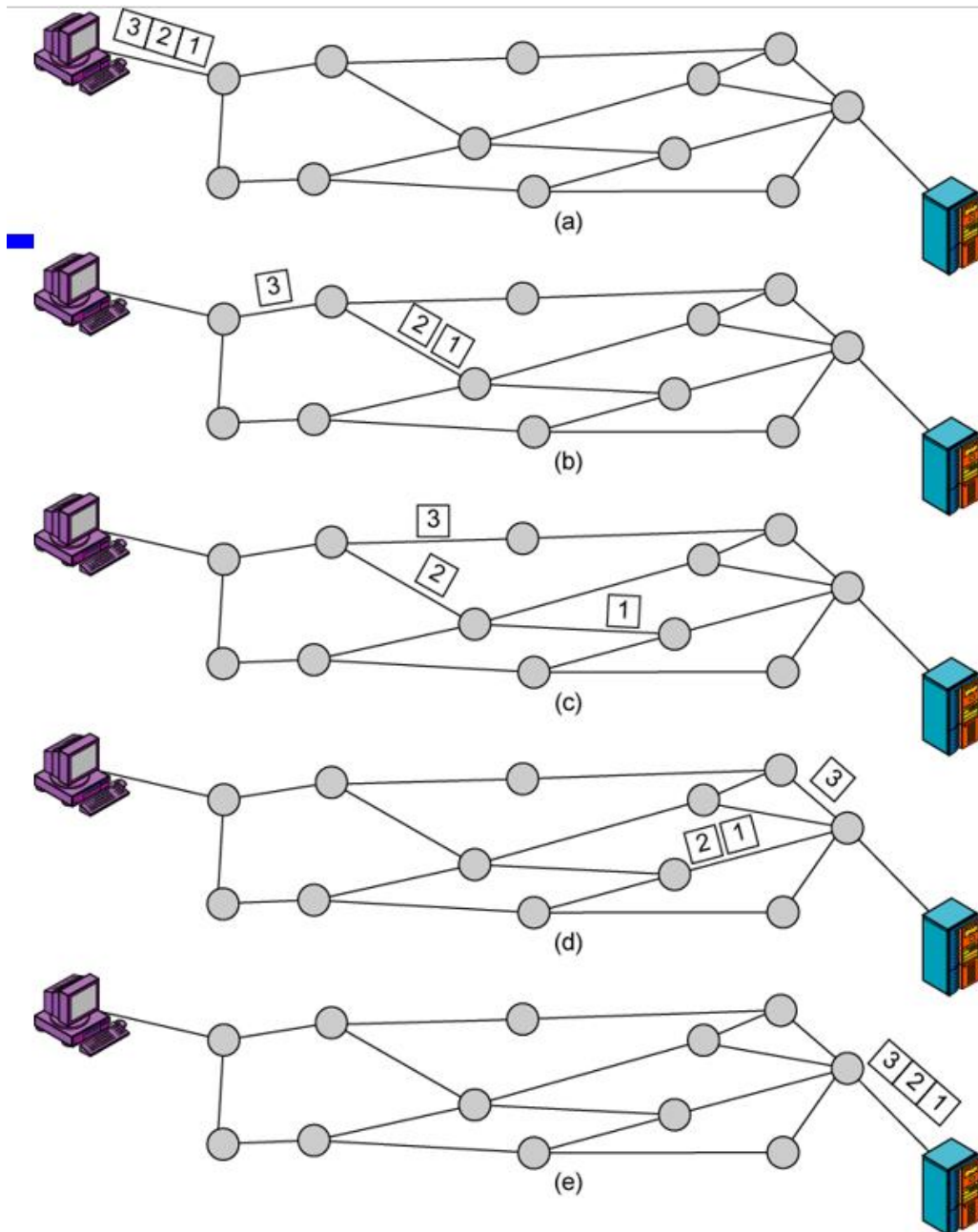
Packet handling

- Packets handled in two ways
 - Datagram
 - Virtual circuit

Datagram

- Each packet treated independently
- Packets can take any practical route
- Packets may arrive out of order
- Packets may go missing
- Up to receiver to re-order packets and recover from missing packets

Datagram



WiFi

- University Wi-Fi networks use **datagram packet switching (IP)** rather than virtual circuits
- Authentication may create the appearance of a session

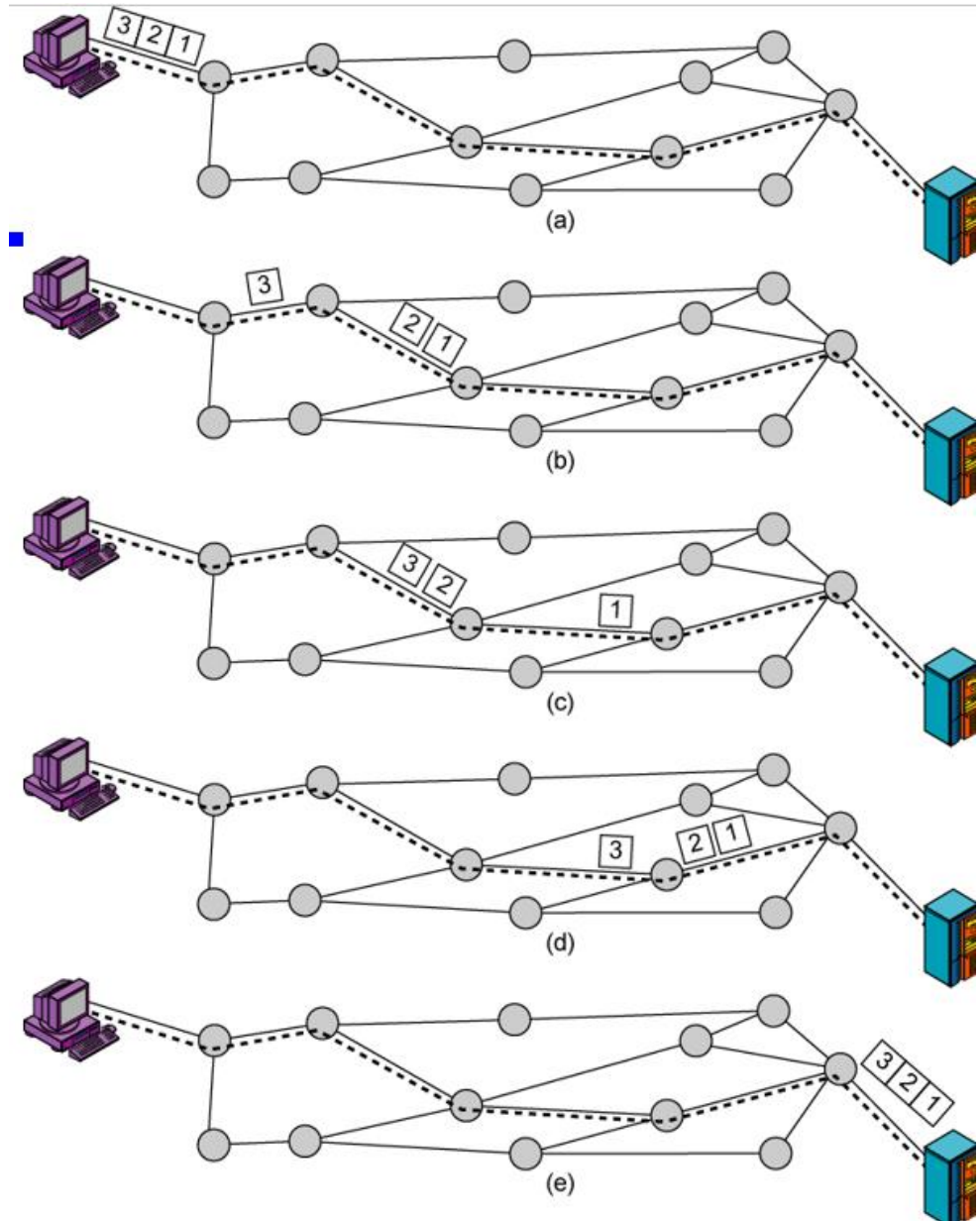
University WiFi

- Devices send **IP packets**
 - Routers forward packets independently
 - No fixed end-to-end path is reserved
 - Paths can change dynamically
-
- → Datagram Network?

Virtual Circuit

- Preplanned route established before any packets are sent
- Call request and call accept packets establish connection (handshake)
- Each packet contains a virtual circuit identifier instead of destination address
- No routing decisions required for each packet
- Clear request to drop circuit
- Not a dedicated path

Virtual Circuit



- Wi-Fi in labs → datagram
 - Ethernet in buildings → datagram
 - ISP MPLS link between campuses → virtual-circuit-like path
-
- Ethernet is a wired LAN technology that enables computers and other devices to send and receive data in the form of frames over network cables
 - MPLS (Multiprotocol Label Switching) → Widely used by ISPs and large enterprises

Difference

- Virtual Circuit is the connection oriented implemented through buffers etc.
- On other hand Datagram is the connection less service where no such resources are required for the data transmission
- A Datagram based network is a true packet switched network. There is no fixed path for transmitting data

Chapter Ref.

- Book-1, Ch 1
- Book-2, Ch 7